Deviance, AIC, and model building

SBA data

Data on loan defaults for US Small Business Administration (SBA) loans:

- `Default`: whether the business defaulted on the loan (1 = yes, 0 = no)
- `UrbanRural`: 1 if business is in urban area, 2 if business is in rural area, 0 if unknown
- `NewExist`: 1 if business already existed, 2 if business is new, 0 if unknown
- `Term`: Length of the loan term (months)
- `DisbursementGross`: The amount of money disbursed (loaned), in dollars
- Many other variables...

Last time: Class activity

```
m1 <- glm(Default ~ log(DisbursementGross) + Term +
            as.factor(UrbanRural),
          family = binomial, data = sba)

table(Prediction = m1$fitted.values > 0.5,
      Truth = sba$Default)
```

```
##            Truth
## Prediction FALSE TRUE
##      FALSE  3989  734
##      TRUE    100  168
```

good at predictions for loans that did not default

bad at predictions for loans that did default

- Accuracy = $(3989 + 168) / 4991 \approx 0.83$
- Sensitivity = $168/(168+734) = 0.19$ ← very low
- Specificity = $3989/(3989+100) \approx 0.98$ ← very high
- PPV = $\frac{168}{168+100} = 0.63$

Last time: Class activity

```
##             Truth
## Prediction FALSE TRUE
##      FALSE  3989  734
##      TRUE    100  168
```

Consider:

	$Y=0$	$Y=1$
$\hat{Y}=0$	4089	902
$\hat{Y}=1$	0	0

(predict $\hat{Y}=0$ for everyone)

> Is an accuracy of around 80% good?

$$\text{accuracy} = \frac{4089}{4991} = 0.82$$

$= P(Y=0)$ in data

- depends on context
- also depends on proportion of $Y=0$ & $Y=1$ in data
- when we have unbalanced classes ($P(Y=0)$ & $P(Y=1)$ are very different) then accuracy is not meaningful by itself (can get high accuracy by predicting all $\hat{Y}=0$ or all $\hat{Y}=1$)

Last time: Class activity

· Accuracy highest when threshold ≈ 0.5

· As threshold ↑, sens ↓, spec ↑
trade-off between sens + spec

Changing thresholds:

```
table(Prediction = m1$fitted.values > 0.3,
      Truth = sba$Default)
```

```
##              Truth
## Prediction FALSE TRUE
##      FALSE  3524  351
##      TRUE    565  551
```

Accuracy: 0.82

Sensitivity = 0.61

Spec. = 0.86

```
table(Prediction = m1$fitted.values > 0.7,
      Truth = sba$Default)
```

```
##              Truth
## Prediction FALSE TRUE
##      FALSE  4089  902
```

Accuracy = 0.82

Sens = 0

Spec = 1

A new research question

- `Default`: whether the business defaulted on the loan (1 = yes, 0 = no)
- `UrbanRural`: 1 if business is in urban area, 2 if business is in rural area, 0 if unknown
- `NewExist`: 1 if business already existed, 2 if business is new, 0 if unknown
- `Term`: Length of the loan term (months)
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- Many other variables...

> **Research question:** Which combination of variables "best" models loan default?

A new research question

> **Research question:** Which combination of variables "best" models loan default?

We need:

- A metric to compare different models
- A way to efficiently search through many different models

Logistic regression

Data: Grad application data

- `admit`: accepted to grad school? (0 = no, 1 = yes)
- `gre`: GRE score
- `gpa`: undergrad GPA
- `rank`: prestige of undergrad institution

$$Admit_i \sim Bernoulli(\pi_i) \quad \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Goal: Summarize how well our model fits the data

> How do we summarize model fit in *linear* regression?

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

$$R^2_{adj} = 1 - \frac{SSE/(\overset{\#\text{obs}}{n} - \overset{\#\text{params}}{p})}{SSTotal/(n-1)}$$

Summarizing linear regression model fit

Example model:

```
linear_mod <- lm(gre ~ gpa, data = grad_app)
summary(linear_mod)
```

```
...
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   192.30      47.92   4.013 7.15e-05 ***
## gpa           116.64      14.05   8.304 1.60e-15 ***
## Residual standard error: 106.8 on 398 degrees of freedom
## Multiple R-squared:  0.1477,    Adjusted R-squared:  0.1455
...
```

R^2 $\qquad\qquad$ R^2_{adj}

Summarizing linear regression model fit

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linear_mod <- lm(gre ~ gpa, data = grad_app)
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```

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...
## Coefficients:
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## Residual standard error: 106.8 on 398 degrees of freedom
## Multiple R-squared:  0.1477,    Adjusted R-squared:  0.1455
...
```

> Is R^2_{adj} appropriate for logistic regression?   No!

we don't use residuals like this in logistic regression

logistic regression: think about likelihood / log-likelihood

Summarizing logistic regression

```
logistic_mod <- glm(admit ~ gpa, data = grad_app,
                    family = binomial)
summary(logistic_mod)
```

```
...
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -4.3576     1.0353  -4.209 2.57e-05 ***
## gpa           1.0511     0.2989   3.517 0.000437 ***
## ---
##     Null deviance: 499.98  on 399  degrees of freedom
## Residual deviance: 486.97  on 398  degrees of freedom
...
```

R reports deviance instead of R^2, R^2_{adj}

Deviance

```
logistic_mod <- glm(admit ~ gpa, data = grad_app,
                    family = binomial)
summary(logistic_mod)
```

```
...
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -4.3576     1.0353  -4.209 2.57e-05 ***
## gpa           1.0511     0.2989   3.517 0.000437 ***
## ---
##     Null deviance: 499.98  on 399  degrees of freedom
## Residual deviance: 486.97  on 398  degrees of freedom
...
```

For logistic regression: deviance = $-2\log(\text{likelihood})$

- larger likelihood $\Rightarrow$ smaller deviance
- want to minimize deviance when estimating coefficients

Deviance

Example:

```
glm(admit ~ gre, family = binomial, data = grad_app)
```

```
...
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual
## Null Deviance:          500
## Residual Deviance: 486.1      AIC: 490.1
...
```

```
glm(admit ~ gpa, family = binomial, data = grad_app)
```

```
...
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual
## Null Deviance:          500
## Residual Deviance: 487      AIC: 491
...
```

has smaller residual deviance

Which predictor (GRE or GPA) gives a model with a better fit?

Comparing models

```
glm(admit ~ gre, family = binomial, data = grad_app)
```

```
...
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual
## Null Deviance:       500
## Residual Deviance: 486.1     AIC: 490.1
...
```

```
glm(admit ~ gre + gpa, family = binomial, data = grad_app)
```

> How will deviance change when I add another variable to the model?

Comparing models

```
glm(admit ~ gre, family = binomial, data = grad_app)
```

```
...
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual
## Null Deviance:        500
## Residual Deviance: 486.1     AIC: 490.1
...
```

```
glm(admit ~ gre + gpa, family = binomial, data = grad_app)
```

```
...
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual
## Null Deviance:        500
## Residual Deviance: 480.3     AIC: 486.3
...
```

Deviance will *never* increase when adding more explanatory variables to a model!

Comparing models

> In linear regression, what quantity did we use to compare models with different numbers of parameters?

$$R^2_{adj} = 1 - \frac{SSE/(n-p)}{SSTotal/(n-1)} \leftarrow \text{penalty for number of parameters}$$

$$AIC = \text{deviance} + \underbrace{2p}_{\text{penalty for extra parameters}}$$

p = # parameters

AIC

$p=3$ $(\beta_0, \beta_1, \beta_2)$

```
glm(admit ~ gre + gpa, family = binomial, data = grad_app)
```

```
...
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual
## Null Deviance:        500
## Residual Deviance: 480.3     AIC: 486.3
...
```

$480.3 + 2(3) = 486.3$

Suppose our model has p parameters (the number of βs, including the intercept). Then the AIC is

$$AIC = \text{deviance} + 2p$$

AIC

$$AIC = \text{deviance} + 2p$$

Model 1:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 Term_i + \beta_2 \log(DisbursementGross_i)$$

```
Residual Deviance: 3974 AIC: 3980
```

Model 2:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 Term_i + \beta_2 Urban_i + \beta_3 Rural_i + \beta_4 Employees_i$$

```
Residual Deviance: 3827 AIC:
```

$3827 + 2(5) = 3837$

> Which model do we prefer, based on AIC?

Back to the research question

> **Research question:** Which combination of variables "best" models loan default?

We need:

- A metric to compare different models
  - Solution: AIC
- A way to efficiently search through many different models

> How should we search many different models?

Some model search algorithms

Best subset selection:

- fit every combination of variables
- choose combo w/ best AIC

Stepwise selection:

- start w/ initial model
- add / remove variables at each step to try and find best AIC

Class activity

https://sta214-f25.github.io/class_activities/ca_11.html

- Work independently or with a neighbor
- Submit HTML on Canvas when done